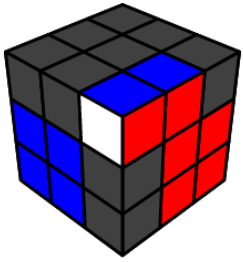
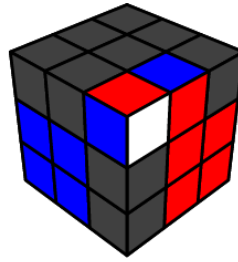


F2L Algorithms



A+
U (R U R')



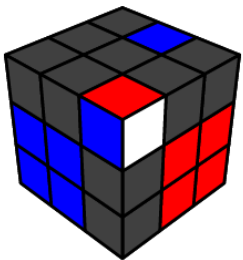
K+
U' (R U' R' U)
R U R'



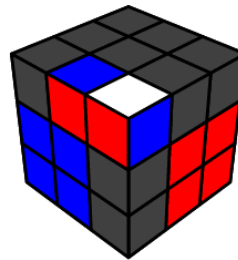
A-
y' U' (R' U R)



K-
d (R' U R U')
R' U' R



B+
R U R'



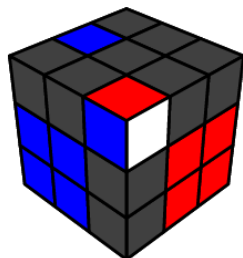
X+
(R U' R' U) (R
U' R' U2 R U'
R')



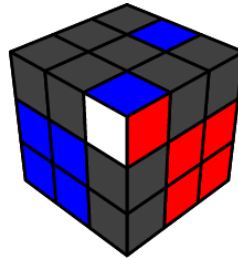
B-
y' (R' U' R)



X-
U' (R U2' R'
d) R' U' R



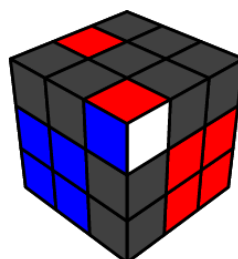
R+
U2 (R U' R'
U') R U R'



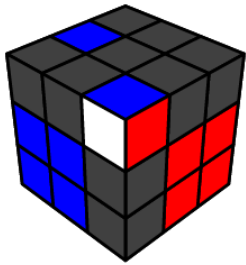
W+
U' (R U R'
U2)
R U' R'



R-
U' (R U' R' d)
R' U' R

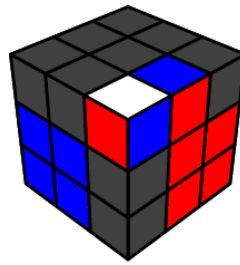


W-
U' r U' (R' U
R U) r'



M+

$U' (R U2' R' U2) R U' R'$



H+

$(R U2' R' U') R U R'$



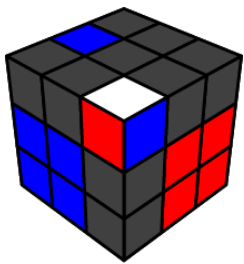
M-

$d (R' U2' R U2) R' U R$



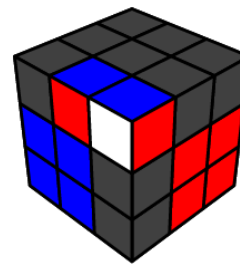
H-

$y' (R' U2' R U) R' U' R$



P+

$U2 (R U R' U) R U' R'$



I+

$(R U R' U2) (R U' R' U R U' R')$



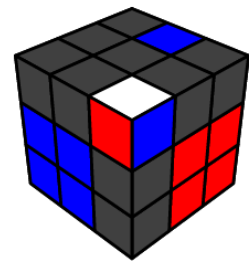
P-

$U2 y' (R' U' R U') R' U R$



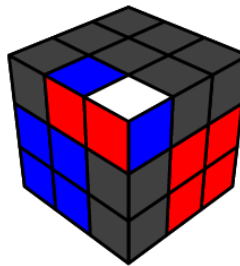
I-

$R U' R' U2 y' R' U' R$



Q+

$U (R U2' R' U) R U' R'$



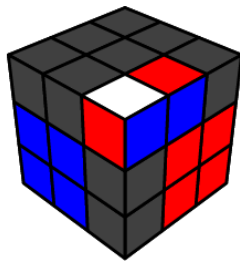
G+

$U (R U' R' U') (R U' R' U R U' R')$



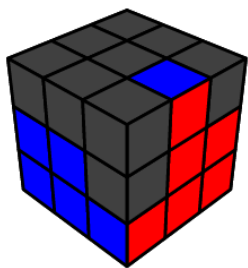
Q-

$y' U' (R' U2 R U') R' U R$

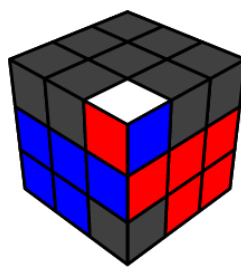


G-

$(F U R U' R' F') R U' R'$



E+
 $U' (R' F R F')$
 $R U R'$



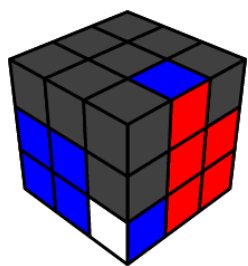
S
 $U (R U' R' U)^2$
 $R U' R'$



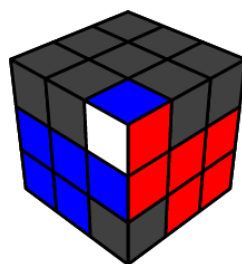
E-
 $U R U' R'$
 $y' U' R' U R$



T
 $R U' R' U y'$
 $R' U R$



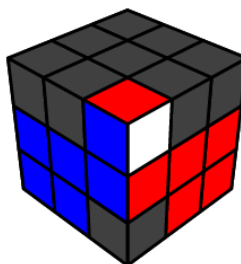
J+
 $(R U' R' U)$
 $R U' R'$



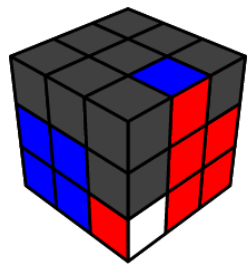
U+
 $U' (R U' R' U^2)$
 $R U' R'$



J-
 $y' (R' U R U')$
 $R' U R$



U-
 $U (R U R' U^2)$
 $R U R'$



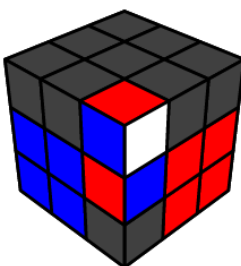
L+
 $(R U R' U')$
 $R U R'$



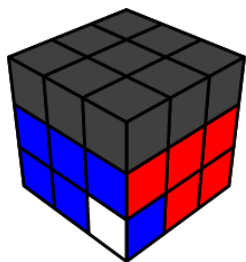
V+
 $U' (R U R' d)$
 $R' U' R$



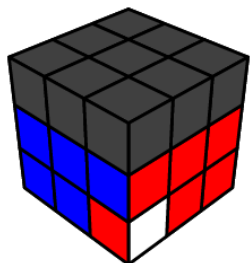
L-
 $y' (R' U' R U)$
 $R' U' R$



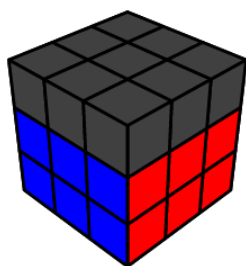
V-
 $U^2 (F' U' F U)$
 $R U' R'$



D+
 (R U' R' U')
 (R U R' U2 R
 U' R')



D-
 (R U' R' U)
 (R U2' R' U
 R U' R')



O
 Solved



C+
 (R U' R' d) (R'
 U' R U' R' U'
 R)

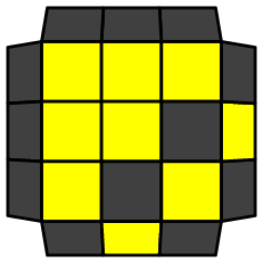


C-
 (R U' R' d) (U
 R' U' R U' R'
 U R)



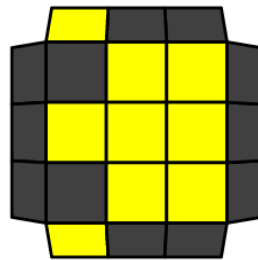
F
 (R U' R' d) (R'
 U2 R U2' R' U R)

OLL Algorithms



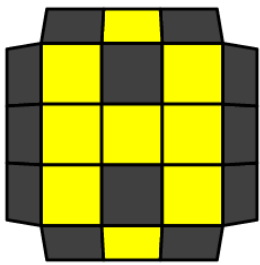
A

$(r\ U\ R'\ U')\ (r'\ R\ U\ R\ U'\ R')$



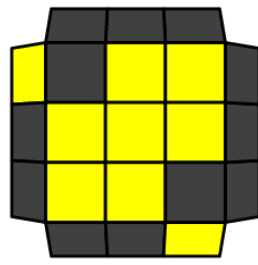
T

$(r\ U\ R'\ U')\ (r'\ F\ R\ F')$



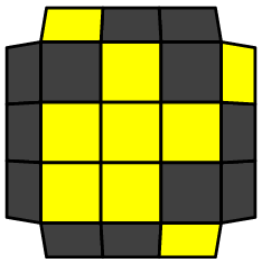
B

$(R\ U\ R'\ U')\ (M'\ U\ R\ U'\ r')$



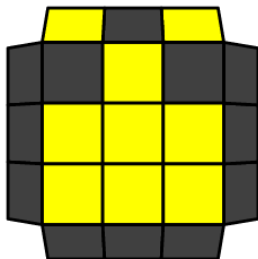
L

$F'\ (r\ U\ R'\ U')\ r'\ F\ R$



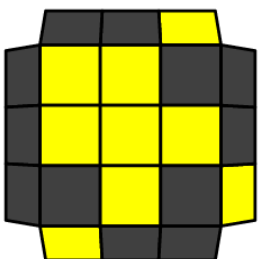
S+

$R\ U\ R'\ U\ R\ U2'\ R'$



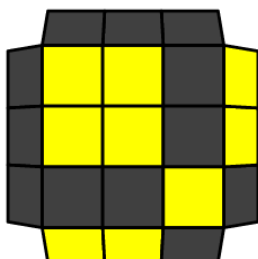
U

$R2'\ D'\ (R\ U2\ R'\ D)\ R\ U2\ R$



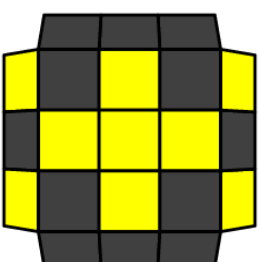
S-

$R'\ U'\ R\ U'\ R'\ U2\ R$



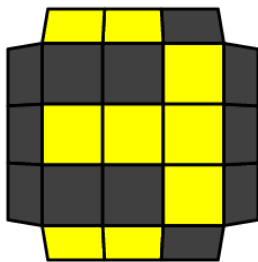
K2

$F\ (R\ U'\ R'\ U')\ (R\ U\ R'\ F')$



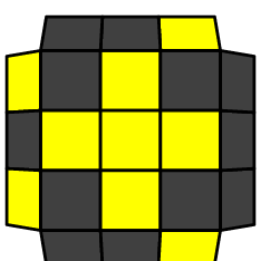
H

$(R\ U\ R'\ U)\ (R\ U'\ R'\ U)\ R\ U2'\ R'$



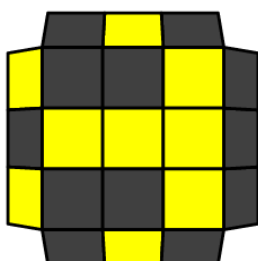
T2

$(R\ U\ R'\ U')\ (R'\ F\ R\ F')$



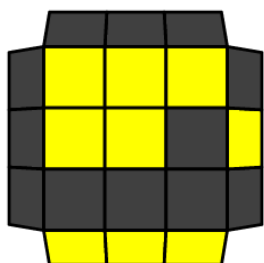
Pi

$R\ U2'\ R2'\ U'\ R2\ U'\ R2'\ U2'\ R$



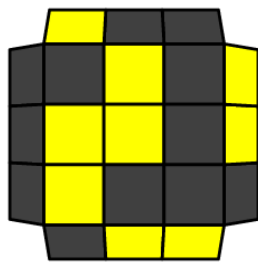
T1

$F\ R\ U\ R'\ U'\ F'$



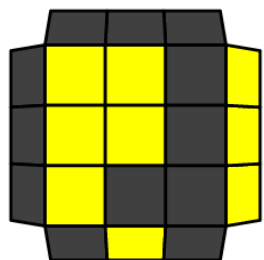
P-

$R' U' F' U F R$



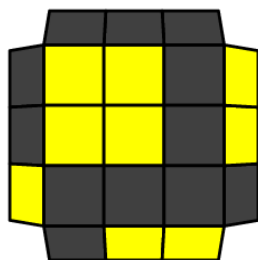
N+

$r U R' U R U 2' r'$



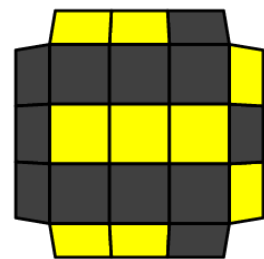
P+

$F (U R U' R') F'$



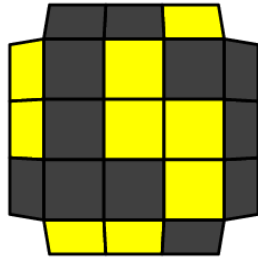
O+

$I' U 2 L U L' U I$



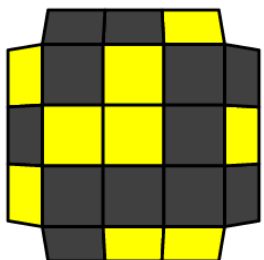
I2

$F (U R U' R') 2 F'$



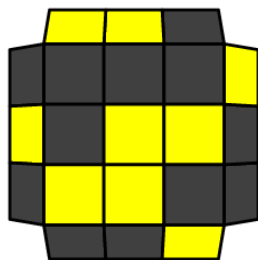
N-

$I' U' L U' L' U 2 I$



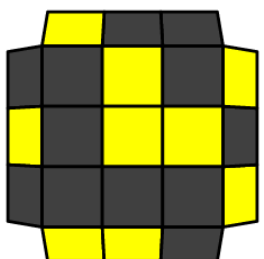
V+

$F (R U R' U') 2 F'$



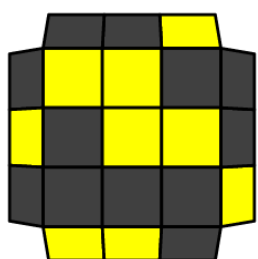
M+

$r' (R 2 U R' U) (R U 2' R' U) M'$



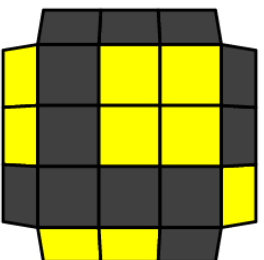
V-

$R' U' (R' F R F') 2 U R$



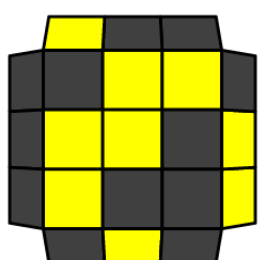
M-

$r (R 2' U' R U') (R' U 2 R U') r' R$



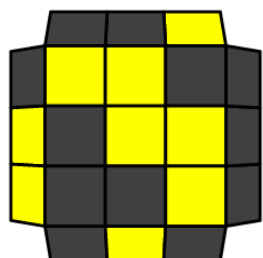
O-

$r U 2' R' U' R U' r'$



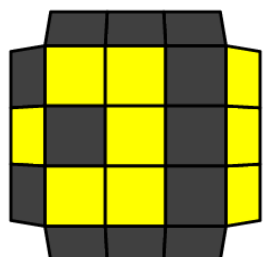
W+

$(R U R' U) (R U' R' U') (R' F R F')$



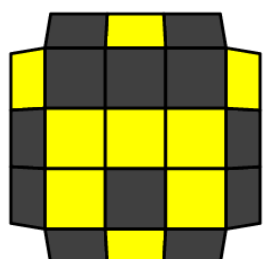
W-

$(L' U' L U') (L' U L U) (L F' L' F)$



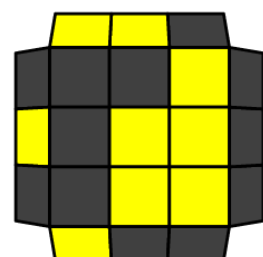
CU

$R' U' (R' F R F') U R$



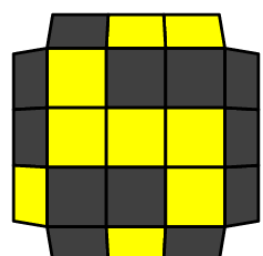
CT

$(R U R^2 U') (R' F R U R U' F')$



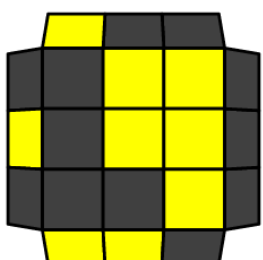
Q+

$S (R U R' U') (R' F R f')$



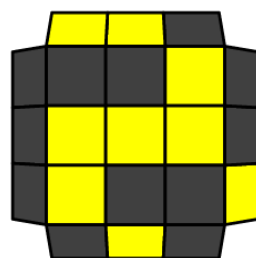
Z+

$R' F (R U R' U') F' U R$



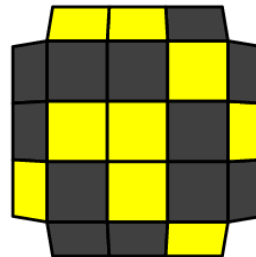
Q-

$R' U' (F U R U' R' F') R$



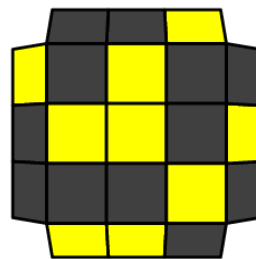
Z-

$L F' (L' U' L U) F U' L'$



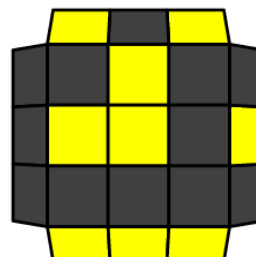
Y+

$(R U R' U) (R' F R F') R U^2 R'$



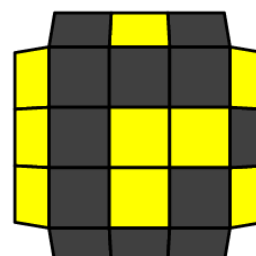
Y-

$(R U R' U') (R' F R^2 U R' U' F')$



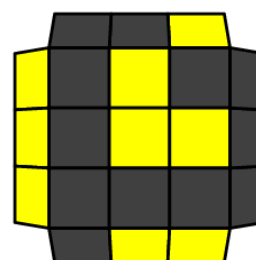
E+

$(r U^2 R' U') (R U R' U') R U' r'$



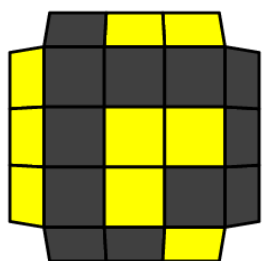
E-

$r' U' (R U' R' U) (R U' R' U^2) r$



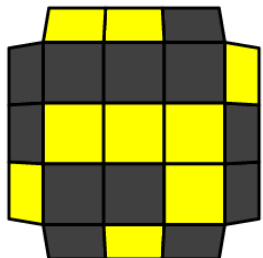
R+

$r U' r^2 U r^2 U r^2 U' r$



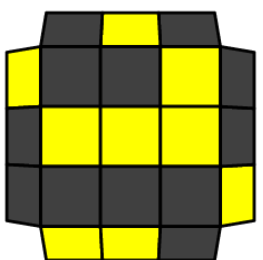
R-

$r' U r^2 U' r^2 U'$
 $r^2 U r'$



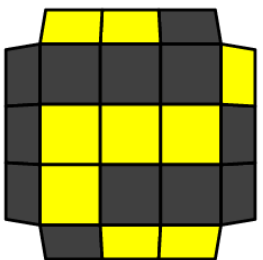
L+

$r' U' r (R' U' R$
 $U) r' U r$



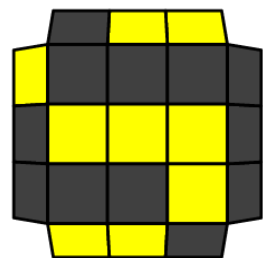
L-

$r U r' (R U R'$
 $U') r U' r'$



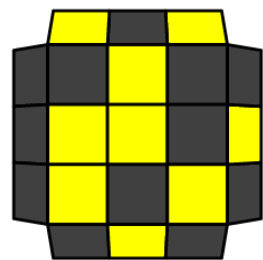
J+

$(r U' r' U') r U r'$
 $y' R' U R$



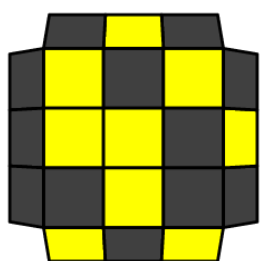
J-

$R' F (R U R' F')$
 $(R F U' F')$



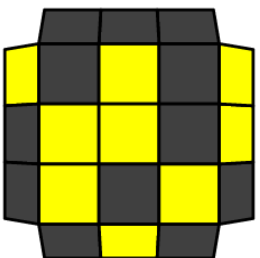
G+

$(R U R' U R$
 $U^2 R') (F R U$
 $R' U' F')$



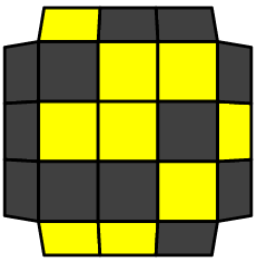
G-

$(R' U' R U' R'$
 $U^2 R) (F R U R'$
 $U' F')$



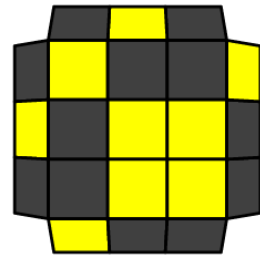
F+

$F U (R U^2 R'$
 $U')^2 F'$



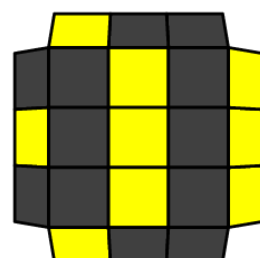
F-

$(R U R' U') (R$
 $U' R') (F' U' F)$
 $(R U R')$



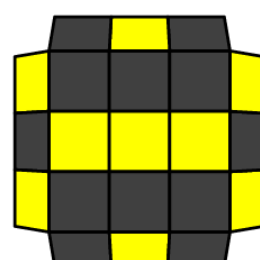
K1

$(R U^2 R^2 F R$
 $F') R U^2 R'$



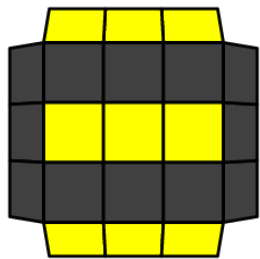
I3

$(R U R' U) (R$
 U'
 $B U' B' R')$



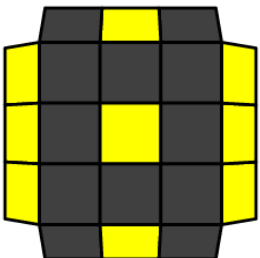
I1

$r U r' (U R U'$
 $R')^2 r U' r'$



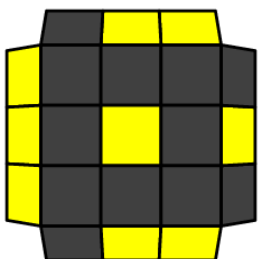
I4

$(R' F R U R U'$
 $R2' F') (R2 U'$
 $R' U R U R')$



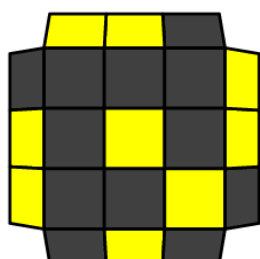
DH

$(R U2' R2' F R$
 $F') U2' (R' F R$
 $F')$



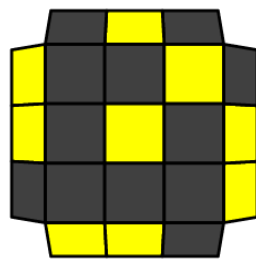
DPi

$(F R U R' U' F')$
 $(f R U R' U' f')$



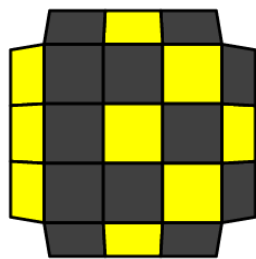
DS+

$(f R U R' U' f')$
 $U' (F R U R' U'$
 $F')$



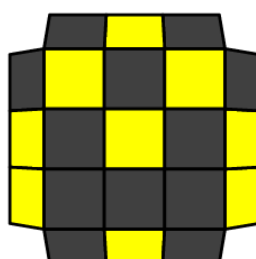
DS-

$(f R U R' U' f')$
 $U (F R U R' U'$
 $F')$



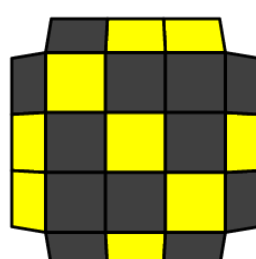
DU

$(R U2' R2' F R$
 $F') U2 (M' U R$
 $U' r')$



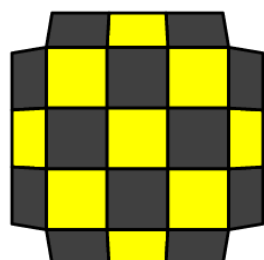
DT

$r' R U (R U R'$
 $U') r (R2' F R$
 $F')$



DL

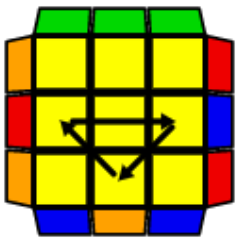
$(R U R' U) (R' F$
 $R F') U2' (R' F$
 $R F')$



X

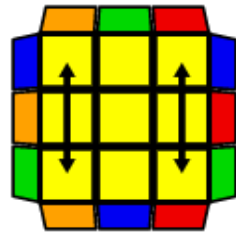
$r' R U (R U R'$
 $U') R'M' (r U$
 R
 $U' r')$

PLL Algorithms



Ub

$R^2 U (R U R' U') R' U' (R' U R')$



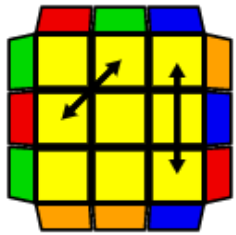
E

$x' (R U' R' D) (R U R' D') (R U R' D) (R U' R' D') x$



Ua

$(R U' R U) R U (R U' R' U') R^2$



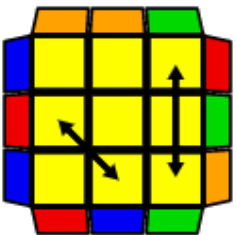
Ra

$(R U' R' U') (R U R D) (R' U' R D') (R' U^2 R')$



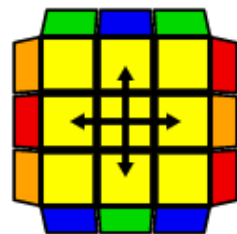
Z

$M' U (M^2 U M^2') U (M' U^2 M^2) [U']$



Rb

$(R' U^2 R U^2') R' F (R U R' U') R' F' R^2 [U']$



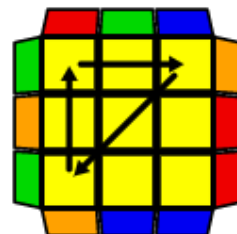
H

$(M^2 U M^2') U^2 (M^2 U M^2')$



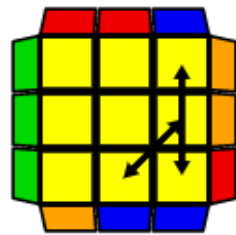
Ja

$(R' U L' U^2) (R U' R' U^2 R) L [U']$



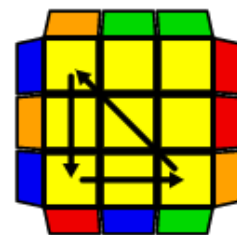
Aa

$x (R' U R') D^2 (R U' R') D^2 R^2 x'$



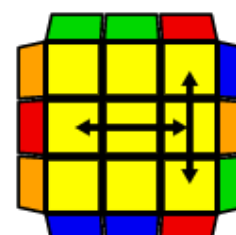
Jb

$(R U R' F') (R U R' U') R' F R^2 U' R' [U']$



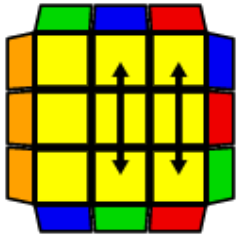
Ab

$x R^2 D^2 (R U R') D^2 (R U' R) x' U^2$



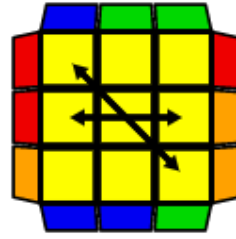
T

$(R U R' U') (R' F R^2 U') R' U' (R U R' F')$



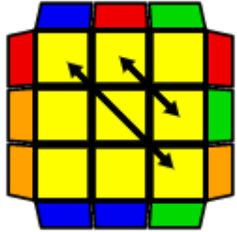
F

$(R' U' F') (R U$
 $R' U') (R' F R^2$
 $U') (R' U' R U)$
 $(R' U R)$



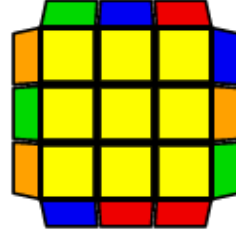
Nb

$(R' U R U') (R' F'$
 $U' F) (R U R' F)$
 $R' F' (R U' R)$



V

$(R' U R' U') y$
 $(R' F' R^2 U') (R'$
 $U R' F) R F$



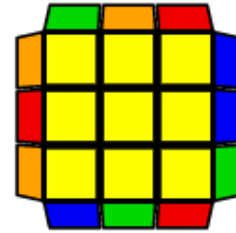
Ga

$R^2 U (R' U R' U')$
 $(R U' R^2) D U'$
 $(R' U R D') [U]$



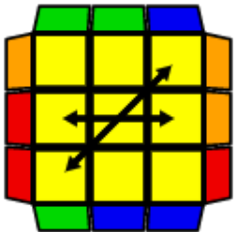
Y

$F (R U' R' U')$
 $(R U R' F') (R$
 $U R' U') (R' F$
 $R F')$



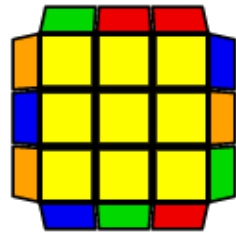
Gb

$y (F' U' F) (R^2 u$
 $R' U) (R U' R u')$
 R^2'



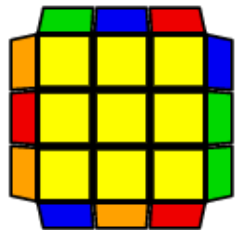
Na

$(R U R' U) (R U$
 $R' F') (R U R'$
 $U') (R' F R^2 U')$
 $R' U^2 (R U' R')$



Gc

$R^2 U' (R U' R U)$
 $(R' U R^2 D') (U R$
 $U' R') D [U']$



Gd

$D' (R U R' U') D$
 $(R^2 U' R U') (R'$
 $U R' U) R^2 [U]$

